

Acceptance Criteria

Given / When / Then format for key user stories

Attraction Discovery (List)

US-01

Attraction list — browse by category

Given a user opens the attraction list — When the page loads — Then all available category filters are displayed (Park, Culture, Landmark, Museum) and the default view shows all categories.

Given a user taps a category filter — When the filter is applied — Then the list updates to show only attractions in that category, and the active filter is visually highlighted.

Given a category is selected — When the user taps the same filter again — Then the filter is removed.

US-02 / US-03

Attraction list with busyness & status

Given a user opens the attraction list — When the list loads — Then each item shows: name, image, rating (0–5), real-time crowd level (very low / low / moderate / busy / very busy), and open/closed status.

Given a user selects a category filter — When filter is applied — Then only attractions of that category are shown.

Given the accessibility filter is ON — When the list renders — Then only mobility-accessible attractions appear and each card shows an accessibility icon.

Given the accessibility filter is ON and a category filter is also active — When both filters are applied — Then results show only attractions matching both the selected category and mobility-accessible criteria.

Attraction Detail

US-04 / US-05

Attraction detail with busyness timeline

Given a user opens an attraction detail page — When the page loads — Then a busyness timeline is shown for today and tomorrow in 30-min intervals, colored by busyness level, and each bar graph shows open/closed status.

Given the timeline is rendered (the detail attraction page) — When a "gem period" exists (low crowd + open + high rating) — Then it is visually highlighted with a gem icon and recommended label.

US-06

Favorite, check-in & rating

Given a logged-in user is on an attraction detail page — When they tap the favorite button — Then the attraction is added to their favorites list, the button state changes to "favorited", and the action is persisted across sessions.

Given a logged-in user is on an attraction detail page — When they tap the check-in button — Then a check-in is recorded with the current timestamp, a new 2.5D gem is added to their gem map, and the check-in appears on their profile map.

Given a logged-in user is on an attraction detail page — When they tap the rating button and submit a score — Then their rating (1–5 stars) is saved, and visible on their profile ratings list.

Gem Recommendation

US-07 / US-08

Gem recommendations

Given a user opens Gem recommendations — When the list loads — Then each entry is: currently open, crowd level very low or low, rating $\geq 75\%$, and no attraction repeats within the same session (each slot is 30min).

Given recommendations are shown — When a time slot has no qualifying gems — Then a fallback message is displayed (e.g. "No quiet gems available for this slot").

Map & Visualization

US-09

Crowd heat map

Given a user opens the map — When the map loads — Then Manhattan zones are displayed as color blocks (5 colors) reflecting current crowd level, with a time slider for today and tomorrow.

Given the slider is moved to a future time — When the map updates — Then zone colors reflect the predicted busyness level for that time.

User System

US-10 / US-11

User system & profile

Given a new user submits the registration form — When email format is valid and password meets requirements — Then account is created with default role "user" and status "active".

Given a logged-in user views their profile — When the page loads — Then favorites list, check-in list (with map pins), and ratings are all displayed with edit and delete options.

Given a user opens the check-in tab — When check-ins exist — Then each check-in is shown as a 2.5D gem on the "gem box" display (on map).

US-12

High-contrast mode

Given a user opens display settings — When they select "High contrast" mode — Then the UI switches to a high-contrast theme (increased text/background contrast ratio) and the selected mode is persisted for future sessions.

Given high-contrast mode is active — When any page is rendered — Then all interactive elements (buttons, links, inputs) remain clearly distinguishable and functional with no loss of information.

Given a user is in high-contrast mode — When they switch back to normal mode — Then the UI returns to the standard theme immediately and the preference is updated.

Admin System

US-13

Admin — user management

Given an admin is on the user management page — When the page loads — Then a page divide list of all users is displayed, showing: user ID, email, role, account status, and registration date. Each page shows a configurable number of records (default 20).

Given an admin views a user record — When they change the user's role (super admin / admin / user) — Then the role is updated immediately, and the affected user's permissions update on their next session.

Given an admin clicks "Disable account" on a user — When the action is confirmed — Then the user's status changes to "disabled" (0), affecting the user's status update on their next action, the user cannot log in until re-enabled.

Given an admin re-enables a disabled account — When the action is confirmed — Then the user's status changes to "active" (1) and they can log in again immediately.

Given an admin is logged in — When they attempt to change the role of another admin — Then the action is blocked and an error message is shown ("Cannot modify another admin's role")(Only one super admin in the system).